EMMA REY SÁNCHEZ

ARTIFICIAL INTELLIGENCE ENGINEER

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github.com/emmareysanchez | linkedin.com/in/emma-rey-sanchez | Portfolio

PROFILE

AI Engineer at Deloitte with a BSc in Mathematical Engineering with Artificial Intelligence, currently pursuing an MSc in Advanced Artificial Intelligence. Background in probability, optimisation and information geometry, with practical work in NLP, computer vision, generative AI and model evaluation.

EXPERIENCE

Artificial Intelligence Engineer | Deloitte Sep 2026 - Present

Barcelona, Spain | Full-time

- Developed a Retrieval-Augmented Generation (RAG) system to support context-informed LLM responses.
- Contributed to computer vision and AI agent projects for the pharmaceutical industry.

EDUCATION

MSc in Advanced Artificial Intelligence Sep 2025 - Dec 2026 (expected)

Universidad Pontificia Comillas ICAI | Madrid, Spain

Selected study: Deep Generative Models, Information Geometry, Probabilistic AI and Deep Reinforcement Learning.

BSc in Mathematical Engineering with Artificial Intelligence Sep 2021 - Jun 2025

Universidad Pontificia Comillas ICAI | Madrid, Spain

Selected study: Machine Learning, Deep Learning, NLP, Computer Vision, Robotics and Reinforcement Learning.

Academic Exchange | University of Miami Jan 2025 - Jun 2025

Engineering and AI, with business coursework | Grade: 9.4/10

SELECTED AI PROJECTS

GNN Rewiring Analysis | GitHub

Compared four graph-rewiring strategies across three datasets to analyse over-squashing and over-smoothing; implemented GAT models and Jacobian, embedding-distance and Dirichlet-energy diagnostics.

Bayesian Sentiment Analysis | GitHub

Co-developed a sentiment classifier combining supervised topic modelling, Bayesian mixture-of-experts and heteroscedastic uncertainty; evaluated calibration, selective prediction and out-of-distribution detection.

Travel Genie AI | GitHub

Developed a ReAct travel-planning agent with a local LLM, external API tools, a FastAPI backend and a streaming React/TypeScript interface.

Image Captioning | GitHub

Implemented an InceptionV3-LSTM image-captioning model with greedy and beam-search decoding; evaluated generated captions using BLEU and CIDEr.

Language Model Efficiency | GitHub

Implemented language-model fine-tuning and IFEval evaluation for a challenge constrained to one GPU and a 24-hour training budget.

TECHNICAL SKILLS

Languages & Libraries: Python, SQL, PyTorch, scikit-learn, Hugging Face, MATLAB, R.

AI: Deep Learning, NLP, Computer Vision, Graph Neural Networks, Reinforcement Learning, LLMs, RAG, AI Agents.

Engineering: MLOps, Docker, CI/CD, FastAPI, React, TypeScript, APIs, MongoDB.

ADDITIONAL INFORMATION

Languages: Spanish (Native), English (Proficient; five years in London), French (B2).

Award: Second Place, Algebra Category, Mathematics Team Contest, Comillas ICAI.

Teaching: Volunteer English teacher, Trujillo, Peru (Jun-Jul 2024); seasonal sailing instructor, Escuela de Vela Socaire (five summer seasons).